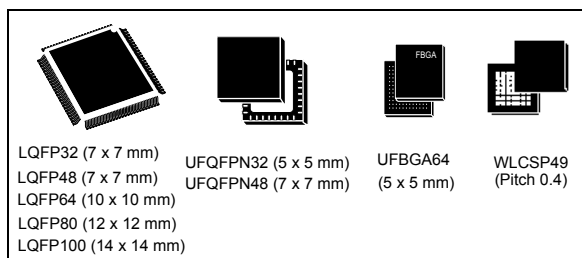


Features

- Core: Arm® 32-bit Cortex®-M4 CPU with FPU, Adaptive real-time accelerator (ART Accelerator) allowing 0-wait-state execution from Flash memory, frequency up to 170 MHz with 213 DMIPS, MPU, DSP instructions
- Operating conditions:
 - VDD, VDDA voltage range: 1.71 V to 3.6 V
- Mathematical hardware accelerators
 - CORDIC for trigonometric functions acceleration
 - FMAC: Filter mathematical accelerator
- Memories
 - 128 Kbytes of Flash memory with ECC support, proprietary code readout protection (PCROP), securable memory area, 1 Kbyte OTP
 - 22 Kbytes of SRAM, with hardware parity check implemented on the first 16 Kbytes
 - Routine booster: 10 Kbytes of SRAM on instruction and data bus, with hardware parity check (CCM SRAM)
- Reset and supply management
 - Power-on/power-down reset (POR/PDR/BOR)
 - Programmable voltage detector (PVD)
 - Low-power modes: sleep, stop, standby and shutdown
 - V_{BAT} supply for RTC and backup registers
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz oscillator with calibration
 - Internal 16 MHz RC with PLL option ($\pm 1\%$)
 - Internal 32 kHz RC oscillator ($\pm 5\%$)
- Up to 86 fast I/Os
 - All mappable on external interrupt vectors
 - Several I/Os with 5 V tolerant capability



- Interconnect matrix
- 12-channel DMA controller
- 2 x ADCs 0.25 μ s (up to 23 channels). Resolution up to 16-bit with hardware oversampling, 0 to 3.6 V conversion range
- 4 x 12-bit DAC channels
 - 2 x buffered external channels 1 MSPS
 - 2 x unbuffered internal channels 15 MSPS
- 4 x ultra-fast rail-to-rail analog comparators
- 3 x operational amplifiers that can be used in PGA mode, all terminals accessible
- Internal voltage reference buffer (VREFBUF) supporting three output voltages (2.048 V, 2.5 V, 2.95 V)
- 14 timers:
 - 1 x 32-bit timer and 2 x 16-bit timers with up to four IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
 - 2 x 16-bit 8-channel advanced motor control timers, with up to 8 x PWM channels, dead time generation and emergency stop
 - 1 x 16-bit timer with 2 x IC/OCs, one OCN/PWM, dead time generation and emergency stop
 - 2 x 16-bit timers with IC/OC/OCN/PWM, dead time generation and emergency stop
 - 2 x watchdog timers (independent, window)
 - 1 x SysTick timer: 24-bit downcounter
 - 2 x 16-bit basic timers
 - 1 x low-power timer

- Calendar RTC with alarm, periodic wakeup from stop/standby
- Communication interfaces
 - 1 x FDCAN controller supporting flexible data rate
 - 3 x I²C Fast mode plus (1 Mbit/s) with 20 mA current sink, SMBus/PMBus, wakeup from stop
 - 4 x USART/UARTs (ISO 7816 interface, LIN, IrDA, modem control)
 - 1 x LPUART
 - 3 x SPIs, 4 to 16 programmable bit frames, 2 x with multiplexed half duplex I²S interface
 - 1 x SAI (serial audio interface)
 - USB 2.0 full-speed interface with LPM and BCD support
 - IRTIM (infrared interface)
 - USB Type-C™ /USB power delivery controller (UCPD)
- True random number generator (RNG)
- CRC calculation unit, 96-bit unique ID
- Development support: serial wire debug (SWD), JTAG, Embedded trace macrocell™

Table 1. Device summary

Reference	Part number
STM32G431x6	STM32G431C6, STM32G431K6, STM32G431R6, STM32G431V6, STM32G431M6
STM32G431x8	STM32G431C8, STM32G431K8, STM32G431R8, STM32G431V8, STM32G431M8
STM32G431xB	STM32G431CB, STM32G431KB, STM32G431RB, STM32G431VB, STM32G431MB